# SOFTWARE REQUIREMENTS SPECIFICATION

IQONIX LABS — Corporate Website

Website for Robotics, Aeronautics, Programming and STEM Education

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| **Document type** | Software Requirements Specification (SRS) |
| **Prepared for** | IQONIX LABS |
| **Intended audience** | Full-stack developer / intern, project owner, QA |
| **Version** | 1.0 — Initial development brief |
| **Date** | 27 October 2026 |

Purpose: Provide a practical, implementation-ready baseline for designing, developing, testing, and launching the IQONIX LABS website. Items marked as recommendations should be confirmed by the project owner before production.

# Document Contents


- 1. Introduction
- 2. Project Overview and Goals
- 3. Users and Target Audience
- 4. Scope
- 5. Sitemap and Page Requirements
- 6. Functional Requirements
- 7. Admin Panel Requirements
- 8. UI/UX and Branding
- 9. Non-Functional Requirements
- 10. Suggested Technical Architecture
- 11. Data Model (Initial)
- 12. Security and Privacy
- 13. Acceptance Criteria
- 14. Deliverables
- 15. Development Phases
- 16. Assumptions and Open Questions

# 1. Introduction

## 1.1 Purpose

This document defines the requirements for the IQONIX LABS corporate website. It is intended to help the assigned full-stack developer build the site with clear expectations, reduce ambiguity, and provide a checklist for testing and handover.

## 1.2 Product Name

**IQONIX LABS Website** — public-facing company website with a content management/admin area.

## 1.3 Business Context

IQONIX LABS provides hands-on STEM education, with a focus on robotics, aeronautics and programming for school students. The website should communicate the value of practical learning to schools and help school decision-makers enquire about programs, demonstrations and lab setups.

## 1.4 Intended Audience

- Project owner / management.
- Full-stack developer or intern responsible for implementation.
- Content editor / administrator.
- Quality assurance tester.

# 2. Project Overview and Goals

Build a professional, responsive, trustworthy website that explains the company's education offerings and converts interested schools into enquiries.

## 2.1 Primary Goals

- Present IQONIX LABS and its educational approach clearly.
- Show robotics, aeronautics, programming and STEM lab solutions for schools.
- Target CBSE and ICSE schools and students in Grades 5–10.
- Generate qualified enquiries for school demonstrations, program discussions and lab setup proposals.
- Display authentic lab activities, student projects, photos and approved outcomes.
- Allow authorized staff to update website content without editing code.

## 2.2 Success Indicators

- Visitors can understand the offering within a short visit and find a clear enquiry call-to-action.
- Enquiry form submissions are stored and visible to authorized administrators.
- Core pages work on mobile, tablet and desktop.
- Administrators can manage programs, gallery items and news/blog content.

Analytics and conversion targets may be added after launch; no numeric target is assumed in this version.

# 3. Users and Target Audience

## 3.1 Website Visitors

- School owners, principals, coordinators and management teams.
- Teachers and STEM / science coordinators.

- Parents and students seeking information about programs.
- Potential partners and other interested organizations.

### 3.2 Admin Users

- Administrator: manages website content, users/roles and enquiries.
- Content editor (recommended): manages approved pages, programs, gallery and blog posts, without access to sensitive settings.

### 3.3 Primary Geographic Focus

Bengaluru / Bangalore and nearby schools initially. The website should not be hard-coded so narrowly that expansion to other locations becomes difficult.

## 4. Scope

### 4.1 In Scope

- Responsive public website.
- Home, About, Programs, STEM Lab Setup, Gallery, Blog/News and Contact pages.
- Enquiry / request-a-demo form.
- Admin login and content management for relevant website sections.
- Enquiry list and enquiry status management.
- Basic SEO, accessibility, performance and security.
- Deployment configuration, documentation and handover.

### 4.2 Out of Scope for Initial Release

- Student learning portal, LMS, online classes or student accounts.
- Online payments, subscriptions or e-commerce.
- Complex CRM integrations, marketing automation or WhatsApp API integration.
- Multi-language website, unless separately approved.
- Native mobile application.

These can be proposed as future enhancements, but should not be added without owner approval.

## 5. Sitemap and Page Requirements

### 5.1 Home

- Clear headline describing hands-on robotics, aeronautics and STEM education.
- Short supporting statement identifying school audience and Grades 5–10.
- Primary CTA: “Request a Demo” or “Discuss a STEM Lab”.
- Introductory sections for programs, school lab setup, practical learning, gallery highlights and contact.
- Use authentic company/lab/student imagery where permission exists.
- Footer with contact information, page links and social links if provided.

### 5.2 About Us

- Company overview, mission, vision and learning philosophy.
- Explain how practical, project-based STEM activities support classroom learning.

- Include company milestones or team information only when verified and approved.

### 5.3 Programs

- List programs for robotics, aeronautics, programming/coding and related STEM learning.
- Each program should support a title, summary, detailed description, suitable grades, learning outcomes, image and CTA.
- Avoid unsupported claims about certifications, guaranteed results or partnerships.

### 5.4 STEM Lab Setup for Schools

- Explain the lab setup offering, such as equipment/kit categories, curriculum support, implementation and ongoing learning activities, based on the actual services approved by the owner.
- Show the process: enquiry, requirement discussion, proposed setup, implementation and support, where applicable.
- CTA to request a consultation or demonstration.

### 5.5 Gallery

- Display approved photos/videos of lab setups, student projects, activities and events.
- Allow categories or filters if practical.
- Images must be optimized and include meaningful alt text.
- Obtain appropriate consent before publishing identifiable student photos or videos.

### 5.6 Blog / News

- List articles, announcements, STEM activity highlights and company updates.
- Each post supports title, slug, cover image, author (optional), publication date, summary and body.
- Support draft/published status and basic SEO metadata.

### 5.7 Contact / Request a Demo

- Show approved company contact details, email, phone and address/service area, if provided.
- Provide a form with: name, school/organization name, role/designation (optional), email, phone, city, interest area, preferred contact method (optional), message, and consent checkbox where needed.
- Required fields should be clearly marked and validated.
- Show success confirmation after a successful submission and a useful error message if it fails.
- Protect the form against spam and duplicate accidental submissions.

### 5.8 Privacy Policy and Terms

Provide a privacy policy explaining what enquiry data is collected, why it is used, who can access it and how a user can request correction/deletion. Terms may be added if the owner/legal adviser approves the content.

## 6. Functional Requirements

### 6.1 Navigation

- All public pages must be reachable from a consistent header/footer navigation.
- Mobile navigation must be usable by touch and keyboard.
- Active page state and clear CTA should be visible.

### 6.2 Enquiry Form

- Validate required fields on client and server.
- Validate email format and phone input sensibly without assuming every visitor has an Indian number.
- Store submitted enquiries with timestamp and status.
- Default status: New. Admin can update status to Contacted, In Progress, Converted or Closed.
- Show a confirmation only after the server confirms successful storage.
- Do not expose enquiry records publicly.

### 6.3 Search and Content Discovery

- Blog/news listing should support pagination or a “Load more” pattern if content grows.
- Each public post and program should have a stable, readable URL/slug.
- A site-wide search is optional for the first release.

### 6.4 Error and Empty States

- Show helpful empty states when no blog posts, programs or gallery items are published.
- Provide a custom not-found page and safe server-error handling.
- Never display stack traces, database credentials or internal system details to visitors.

## 7. Admin Panel Requirements

### 7.1 Authentication and Roles

- Admin panel must be protected by login.
- Passwords must be securely hashed; never store plaintext passwords.
- Use server-side authorization for every protected operation.
- Recommended roles: Admin and Editor. Admin manages users/settings; Editor manages approved content and enquiries as permitted.
- Provide secure logout and session/token expiry.

### 7.2 Dashboard

- Show summary counts for new enquiries, enquiries by status, published programs and published posts.
- Show recent enquiries with date, name, school/organization, interest area and status.

### 7.3 Page and Program Management

- Create, view, edit, publish/unpublish and delete content, with confirmation before destructive actions.
- Support title, body/description, image, SEO title/description, slug and publication status where applicable.
- Prevent duplicate slugs or handle them safely.

### 7.4 Gallery Management

- Upload, edit caption/alt text, assign category, publish/unpublish and delete images.
- Restrict file types and sizes; validate uploads on the server.

### 7.5 Blog / News Management

- Create and edit posts, save drafts, publish/unpublish, set cover image and publication date.
- Sanitize rich text or HTML to prevent cross-site scripting.

### 7.6 Enquiry Management

- View and search/filter enquiries by date, status, city or interest area.
- Open a detail view and update status or internal notes.
- Only authorized users may view personal contact data.
- CSV export is optional and must be permission-controlled.

### 7.7 Admin Audit (Recommended)

Record important actions such as content publication, user/role changes and enquiry status updates, including actor and timestamp.

## 8. UI/UX and Branding

### 8.1 Visual Direction

- Professional, modern, education-focused and trustworthy.
- Primary palette: blue and white, with subtle orange accents if suitable.
- Use consistent typography, spacing, buttons, cards and image treatment.
- Avoid clutter, excessive animation and generic template-like sections.

### 8.2 Responsive Design

- Mobile-first layout, then tablet and desktop breakpoints.
- No horizontal overflow at common viewport sizes.
- Forms, navigation and CTA buttons must remain usable on touch screens.

### 8.3 Content and Images

- Prefer genuine IQONIX LABS lab photos, equipment, student projects and approved team images.
- Do not fabricate school partnerships, testimonials, awards, student results or certifications.
- Use stock images only if approved and clearly appropriate.

### 8.4 Accessibility

- Use semantic headings and landmarks.
- Ensure keyboard access, visible focus states, descriptive labels and meaningful alt text.
- Maintain readable color contrast and do not communicate status by color alone.

## 9. Non-Functional Requirements

### 9.1 Performance

- Optimize images, lazy-load below-the-fold media and minimize unnecessary JavaScript.
- Use caching and compression where supported by hosting.
- Target good Core Web Vitals; establish measurable thresholds during implementation and test on representative devices/networks.

### 9.2 SEO

- Unique page titles and meta descriptions.
- Readable URLs, canonical URLs where needed, sitemap.xml and robots.txt.

- Open Graph metadata for social sharing.
- Structured data only where accurate and relevant.
- Use descriptive image alt text and proper heading hierarchy.

### 9.3 Reliability and Maintainability

- Use reusable components and a clear separation of UI, API and data access.
- Keep configuration and secrets in environment variables.
- Provide understandable error logging without leaking sensitive data.
- Document setup, deployment, backup and recovery steps.

### 9.4 Browser Support

Support current stable versions of Chrome, Edge, Firefox and Safari, plus common Android and iOS mobile browsers.

### 9.5 Scalability

The site should allow additional programs, gallery items, posts and enquiry volume without requiring structural redesign.

## 10. Suggested Technical Architecture

The exact stack may be confirmed by the project owner. The following is a recommendation, not a mandatory requirement.

- **Frontend:** React with a framework suitable for SEO (Next.js is recommended) or another approved solution.
- **Backend:** Node.js with Express/NestJS, or server-side framework features provided by the chosen web framework.
- **Database:** PostgreSQL or MongoDB; choose one based on developer expertise and hosting requirements. Avoid introducing multiple databases without a clear reason.
- **Media storage:** Object storage or a managed media service for uploaded images; avoid storing large image binaries directly in the main database.
- **Deployment:** Approved hosting with HTTPS, environment configuration, logs and backups.
- **Version control:** Git repository with a README and clear development/deployment instructions.

### 10.1 Suggested Environments

- Development: local environment with sample data.
- Staging (recommended): preview environment for owner review.
- Production: public domain with HTTPS and production secrets.

### 10.2 API Outline (if a separate API is used)

- Public: GET published programs, gallery items, blog posts and POST enquiry.
- Admin: authenticated CRUD endpoints for programs, gallery, posts and enquiries.
- Use consistent response formats, validation, pagination and appropriate HTTP status codes.

## 11. Data Model (Initial)

The developer may refine fields, but must preserve the functional requirements.

### 11.1 Enquiry

- id; name; organization/school; role; email; phone; city; interest_area; preferred_contact_method; message; consent (where required); status; internal_notes; created_at; updated_at.

### 11.2 Program

- id; title; slug; short_description; full_description; grades; learning_outcomes; image/media reference; SEO title; SEO description; status; created_at; updated_at.

### 11.3 Blog Post

- id; title; slug; summary; body; cover_image; author (optional); published_at; status; SEO title; SEO description; created_at; updated_at.

### 11.4 Gallery Item

- id; media reference; caption; alt_text; category; status; created_at; updated_at.

### 11.5 Admin User

- id; name; email/username; password_hash or identity-provider reference; role; active status; last_login; created_at; updated_at.

Use database constraints and indexes for unique slugs, common filters and timestamps where appropriate.

## 12. Security and Privacy

- Enforce HTTPS in production.
- Validate and sanitize all user input on the server.
- Use parameterized queries or safe ORM patterns.
- Protect against common web risks, including XSS, CSRF where relevant, injection, brute-force login attempts and insecure direct object references.
- Apply rate limiting and spam protection to public enquiry submissions and login.
- Restrict admin access and uploaded file types; store uploads safely.
- Never commit secrets, tokens or passwords to version control.
- Collect only necessary personal information and limit access to it.
- Use backups and document restore steps.
- Student photos/videos must only be published with appropriate school/guardian consent and company approval.
- Define a reasonable retention and deletion process for enquiry data with the owner.

## 13. Acceptance Criteria

- All agreed public pages are implemented and linked through navigation.
- Website layout works at mobile, tablet and desktop sizes.
- Request-a-demo form validates input, saves submissions and shows success/error states correctly.
- Admin login and server-side role checks protect all admin routes and data.
- Authorized admins can create/edit/publish/unpublish programs, gallery items and blog/news posts.
- Admins can view enquiries and update their status.
- No private enquiry data is available to unauthenticated visitors.
- Basic SEO metadata, sitemap and robots.txt are present.

- Images are optimized and accessibility basics are checked.
- No known critical security issues remain at handover.
- README includes setup, environment variables, database setup, run/build/test and deployment instructions.
- Owner has reviewed and approved the staging build before production deployment.

## 14. Deliverables

- Complete responsive website source code in the agreed Git repository.
- Public pages and working enquiry form.
- Admin panel and content management features listed in scope.
- Database schema/migrations or documented data setup.
- Environment variable example file without real secrets.
- README with local setup, commands, deployment and backup/restore notes.
- Basic testing checklist and known-issues list.
- Deployment to the owner-approved hosting/domain, subject to access being provided.
- Short handover walkthrough for the owner.

## 15. Development Phases

| Phase | Work | Output |
|---|---|---|
| 1. Confirm | Confirm content, logo, contacts, stack and hosting. | Approved scope and assets list. |
| 2. Design | Sitemap, wireframes and responsive visual direction. | Owner-approved design. |
| 3. Build public site | Navigation, pages, programs, gallery, blog and forms. | Working public website. |
| 4. Build admin | Authentication, content management and enquiries. | Protected admin panel. |
| 5. Test | Responsive, functional, security and SEO checks. | Resolved issues and test notes. |
| 6. Launch | Deploy, configure HTTPS, verify backups and hand over. | Production release and documentation. |

## 16. Assumptions and Open Questions

The developer can begin with the agreed baseline below, while the project owner confirms outstanding details before production.

- Confirm the final company name and logo assets. This document uses “IQONIX LABS” as supplied in the project discussion.
- Provide approved company description, mission/vision, contact details, social links and legal-policy text.
- Provide genuine lab/student/project photos and confirm permission to publish them.
- Confirm exact programs, curriculum details, grades and what is included in a STEM lab setup.
- Confirm preferred technology stack, database, hosting provider, domain and deployment access.
- Confirm who will administer the website and whether Editor role is needed at launch.

- Confirm whether email notifications should be sent for new enquiries. Email delivery is recommended but depends on a configured provider.
- Confirm whether enquiry CSV export and site-wide search are required for version 1.
- Confirm content ownership, privacy policy and data retention approach.

Until these items are confirmed, use clearly marked placeholder content in development and do not publish unverified claims.

End of SRS — IQONIX LABS Website, Version 1.0